Kodaira embedding theorem

Daniel González Casanova Azuela

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Projective space has a positive form

A point of $\mathbb{P}^n$ is a line through the origin in $\mathbb{C}^{n+1}$. The map $\pi\colon \mathbb{C}^{n+1} \setminus \{0\} \to \mathbb{P}^n$ sends each nonzero vector to the line it spans.

On nonzero vectors $Z \in \mathbb{C}^{n+1}$, define

$$\tilde{\omega} = \frac{i}{2\pi}\,\partial\bar{\partial}\log|Z|^2, \qquad |Z|^2 = |Z_0|^2 + \cdots + |Z_n|^2.$$

It is **basic**: for every vector field V along the fibres of π,

$$\iota_V\tilde{\omega} = 0, \qquad \mathcal{L}_V\tilde{\omega} = 0.$$

So it comes from a form downstairs, the **Fubini–Study form**: $\tilde{\omega} = \pi^*\omega_{\mathrm{FS}}$.

ω_{FS} is **positive**: it gives every nonzero tangent vector a positive squared length.

Positivity passes to submanifolds

Let $X \subset \mathbb{P}^N$ be a compact complex submanifold.

Restrict the line bundle: $L = \mathcal{O}(1)|_X$.

Its curvature is $\omega_{\mathrm{FS}}|_X$, and it is still positive.

$$X \text{ projective } \implies X \text{ has a positive line bundle.}$$

Question: is the converse true?

Kodaira's embedding theorem (1954)

Let X be compact complex and L a holomorphic line bundle with a smooth positive metric. For all sufficiently large N,

$$\boxed{\Phi_{L^N} : X \hookrightarrow \mathbb{P}\big(H^0(X, L^N)^*\big)}$$

is a holomorphic embedding.

Kunihiko Kodaira (1915–1997)

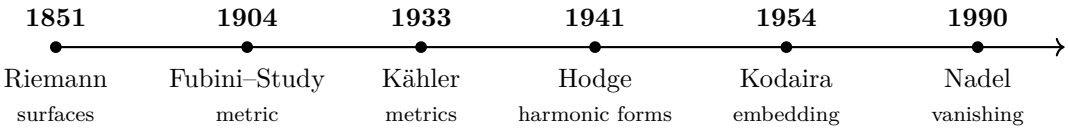


Fields Medal, 1954, for **harmonic integrals** and their applications to Kähler and algebraic varieties.

His idea: an **analytic** condition (curvature) produces an **algebraic** object (polynomial equations).

Photo: Konrad Jacobs / MFO, Tokyo 1969, CC BY-SA 2.0 DE. Historical source: IAS scholar profile. Fields citation: IMU.

Where this sits in history



One theme: **analysis proves that enough holomorphic functions exist**.

Riemann used analysis to prove that compact Riemann surfaces are algebraic curves. Kodaira did the same in every dimension.

An embedding needs values and derivatives

$$\Phi_{L^N}: \begin{array}{ccc} X & \longrightarrow & \mathbb{P}(H^0(X, L^N)^*) \\ x & \longmapsto & [s_0(x) : \cdots : s_M(x)] \end{array} \qquad s_0, \ldots, s_M \text{ a basis of } H^0(X, L^N).$$

Φ_{L^N} is	Language of sections
Defined everywhere	**Base-point free**: no common zero
Injective	**Separates points**: independent values
An immersion	**Separates tangent directions**

Vanishing lifts prescribed data

Put $A = L^N$. For an ideal sheaf I,

$$0 \longrightarrow A \otimes I \longrightarrow A \longrightarrow A \otimes (\mathcal{O}_X/I) \longrightarrow 0.$$

The long exact sequence contains

$$H^0(X, A) \longrightarrow H^0(X, A \otimes \mathcal{O}_X/I) \longrightarrow H^1(X, A \otimes I).$$

$$\boxed{H^1(X, A \otimes I) = 0 \quad \Longrightarrow \quad \text{every prescribed quotient section lifts.}}$$

The task is a vanishing theorem

We want, for one sufficiently large N, $H^1(X, L^N \otimes I) = 0$ for three ideals:

$I = \mathfrak{m}_x$	gives	base-point freeness
$I = \mathfrak{m}_x \cap \mathfrak{m}_y$	gives	point separation
$I = \mathfrak{m}_x^2$	gives	tangent separation

Kodaira vanishing handles a positive bundle with a canonical twist. We need a theorem that also allows these ideals.

Kodaira vanishing (1953)

Let X be a compact complex manifold, and B a line bundle with a smooth metric of **positive** curvature, $i\Theta_B > 0$.

$$\boxed{H^q(X, K_X \otimes B) = 0 \qquad (q > 0).}$$

Then $\omega = i\Theta_B$ is a Kähler form, so X is Kähler. By compactness, $i\Theta_B \geq \varepsilon\omega$ for any Kähler form ω and some $\varepsilon > 0$.

For us: $B = L^N \otimes K_X^{-1}$ is positive for $N \gg 0$, so $H^q(X, L^N) = 0$. But there is no ideal I yet.

Kodaira, Proc. Nat. Acad. Sci. 39 (1953), 1268–1273.

How the proof goes: Bochner–Kodaira

Every class in $H^q(X, K_X \otimes B)$ has a **harmonic** representative u (Hodge).

Bochner–Kodaira: $\Delta_{\bar\partial} = \Delta_\partial + [\, i\Theta_B \wedge, \Lambda\,]$, and the last term is $\geq q\varepsilon$ on (n,q)-forms if $i\Theta_B \geq \varepsilon\omega$.

Pair with u, integrate over X, and integrate by parts in the Δ_∂ term:

$$\begin{aligned} 0 = \int_X \langle \Delta_{\bar\partial} u, u\rangle \, dV &= \int_X \langle \Delta_\partial u, u\rangle \, dV + \int_X \langle [\, i\Theta_B \wedge, \Lambda\,] u, u\rangle \, dV \\ &= \underbrace{\int_X \left(|\partial u|^2 + |\partial^* u|^2\right) dV}_{\geq 0} + \underbrace{\int_X \langle [\, i\Theta_B \wedge, \Lambda\,] u, u\rangle \, dV}_{\geq q\varepsilon \int_X |u|^2 dV}. \end{aligned}$$

So $q\varepsilon \int_X |u|^2 \, dV \leq 0$; for $q > 0$, $u = 0$.

We will use a similar argument for **singular** metrics.

Kodaira (1953), after Bochner (1946). Singular h: Hörmander's L^2 estimates (1965).

The solution: Nadel vanishing (1990)

Let X be compact Kähler, and B a line bundle with a **singular** metric h, $i\Theta_h \geq \varepsilon\omega$. The **multiplier ideal** keeps the germs that stay square-integrable:

$$\mathcal{I}(h)_x = \{f : \ |f|_h^2 \text{ is integrable near } x\}.$$

$$\boxed{H^q\big(X, K_X \otimes B \otimes \mathcal{I}(h)\big) = 0 \qquad (q > 0).}$$

Take $B = L^N \otimes K_X^{-1}$, so $K_X \otimes B = L^N$. Different choices of singular metric, which we construct in the proof, make the multiplier ideal $\mathfrak{m}_x$, $\mathfrak{m}_x \cap \mathfrak{m}_y$ or $\mathfrak{m}_x^2$.

Nadel, Ann. of Math. 132 (1990).